Lightweight Remote Sensing Change Detection With Progressive Feature Aggregation and Supervised Attention

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***Abstract*— Remote sensing change detection (RSCD) aims to explore surface changes from co-registered pair of images. However, the high cost of memory and computation in previous convolutional neural network (CNN)-based methods prevent their successes from being applied to real-world applications. Therefore, we propose a novel lightweight network, which identifies changes based on the features extracted by mobile networks via progressive feature aggregation and supervised attention, termed as A2Net. Considering the less powerful representation capability of mobile networks, we design a neighbor aggregation module (NAM) to fuse features within nearby stages of the backbone to strengthen the representation capability of temporal features. Then, we propose a progressive change identifying module (PCIM) to extract temporal difference information from bitemporal features. Besides, we design a supervised attention module (SAM) to reweight features for effectively aggregating multilevel features from high levels to low levels. With NAM, PCIM, and SAM incorporated, A2Net can achieve favorable results compared with the state-of-the-art methods on three challenging RSCD datasets with fewer parameters (3.78 M) and lower computation costs (6.02 G). The demo code of this work is publicly available at https://github.com/guanyuezhen/A2Net.**

***Index Terms*— Lightweight remote sensing change detection (RSCD), neighbor feature aggregation, progressive change identifying (PCI), supervised attention.**

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I. Introduction

REMOTE sensing change detection (RSCD) aims to locate and segment the surface changes from the co-registered pair of images captured at different times in the same regions. It is an important problem in the remote sensing sense understanding task and serves as one key step for many real-world tasks, such as global resources monitoring [1], land-use change detecting [2], [3], damage assessment [4], [5], and urban management [6], [7].

Early traditional RSCD methods mainly use hand-crafted features to obtain change results, in which the hand-crafted features usually lead limited detection accuracy. Recently, deep convolutional neural networks (CNNs), with its great feature learning capability, have attracted intensive attention in many computer vision tasks, e.g., semantic segmentation [8], [9], [10], [11], object detection [12], [13], and image retrieval [14]. Nowadays, many change detection methods have been proposed to leverage the powerful representation capabilities of CNNs to address the change detection task and achieved better performance than the traditional approaches [15]. However, we observe that their high detection accuracy is often benefited from the expense of high computation costs and large model sizes.

In Fig. 1, we show the detection accuracy in terms of F1-score, model parameters (Params), and computation costs (FLOPs) of some recently proposed RSCD methods, including STANet [6], L-Unet [16], SNUNet [17], DSIFN [18], BIT [19], and TFI-GR [20]. As seen from the results, although these RSCD methods show promising detection performance, some of them contain amazing model parameters and cost large computational overhead. For example, the DSIFN method costs about 50.71 M parameters and 164.557 G FLOPs to achieve 0.9135 detection accuracy on the LEVIR dataset. Such a situation has inevitably prevented these methods from being applied to some real-world applications, especially for deploying the advanced RSCD methods on unmanned aerial vehicles (UAVs) or satellites in the real world, with limited energy overhead and computational capability. Therefore, there is an essential need to design a lightweight method with fewer model parameters, lower computation costs, and promising detection accuracy for the RSCD task.

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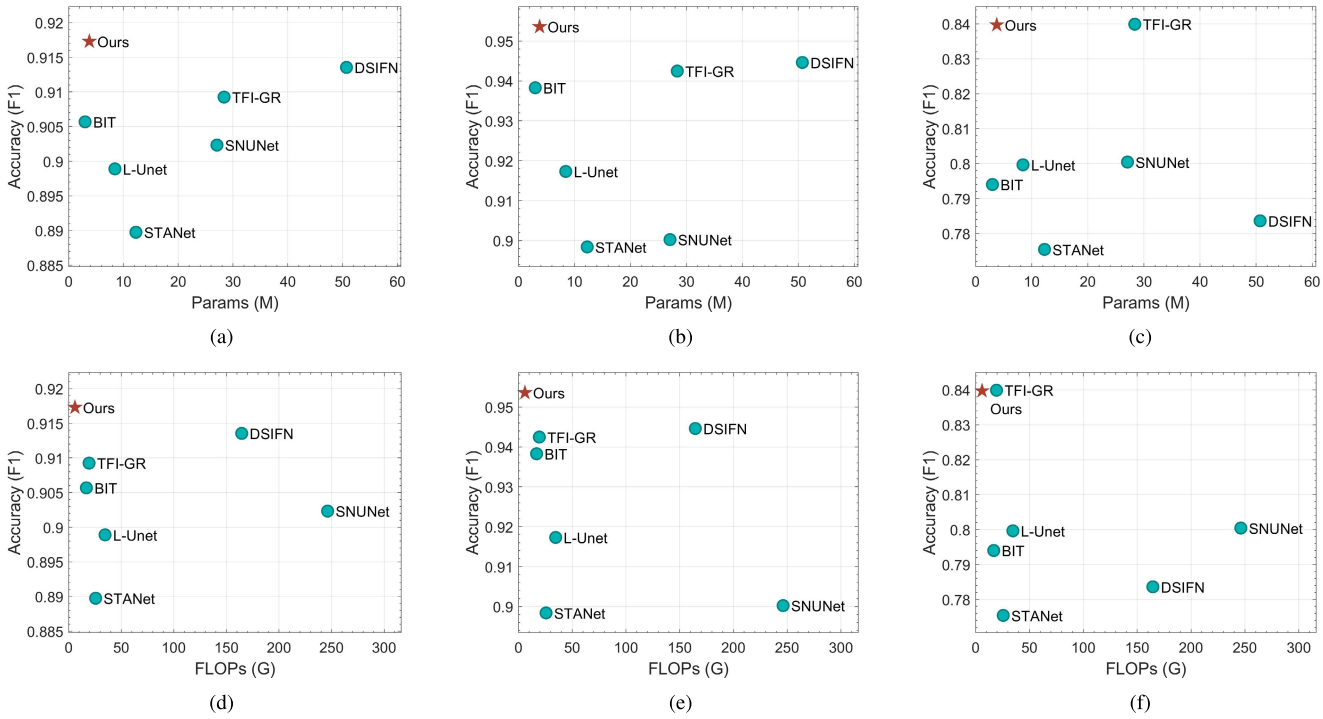


Fig. 1. Model complexity comparison of different methods in terms of Params (memory cost), FLOPs (computational cost), and F1-score on the LEVIR, BCDD, and SYSU datasets, respectively. (a) and (d) LEVIR dataset; (b) and (e) BCDD dataset; and (c) and (f) SYSU dataset. Our proposed A2Net method shows superior accuracy with fewer parameters and lower computation costs.

Based on the above observation, we propose a novel lightweight RSCD network with progressive aggregation and supervised attention, called A2Net. It realizes the lightweight solution from the following two aspects. 1) *Extracting bitemporal features with a lightweight encoder*: As a lightweight context encoder, MobileNetV2 [21] is widely used to obtain pyramid features in many computer version tasks, e.g., image segmentation [22] and object detection [23]. Inspired by the efficiency and effectiveness of MobileNetV2, we use it to extract features from bitemporal images rather than some commonly used cumbersome backbones, i.e., VGG19 [24] and ResNet50 [25]. Considering the lower powerful feature representation capability of MobileNetV2, we construct a neighbor aggregation module (NAM) to fuse the features within nearby stages of backbones. As a result, the semantic information and fine-grained details of objects are further strengthened in the high-level and low-level feature maps, respectively. 2) *Designing a decoder equipped with some lightweight modules*: We implement the decoder in a coarse-to-fine scheme [26] with two key modules, i.e., the progressive change identifying module (PCIM) and the supervised attention module (SAM). We leverage PCIM to efficiently explore the temporal change information from bitemporal features at diverse feature levels. SAM is plugged between every high-level to low-level feature fusion, which efficiently recalibrates multilevel features to enable progressive learning.

With NAM, PCIM, and SAM working together, our A2Net can achieve competitive detection accuracy, e.g., 91.73%, 91.73%, and 91.73% in F1-score on the LEVIR [6], BCDD, and SYSU datasets [27], with fewer parameters (3.78 M parameters) and lower computation costs (6.02 G FLOPs). The main contributions of this work are summarized as follows.

1) We propose a novel lightweight RSCD model (only containing 3.78 M parameters and costing 6.02 G FLOPs), termed A2Net, which progressively aggregates multilevel features with supervised attention guidance to identify temporal changes in a coarse-to-fine strategy.
2) To enhance the representation capability of bitemporal features extracted by the backbone, we introduce an NAM to merge the information of features within nearby stages of the backbone.
3) We propose a PCIM to gradually explore the temporal change information at diverse feature levels for accurately identifying the changed objects. Furthermore, we propose an SAM to reweight features to enable progressive feature aggregation.
4) We conduct a series of experiments to verify the effectiveness and superiority of the proposed method. The experimental results on three benchmark datasets prove that our A2Net achieves better or more competitive performance compared with eight state-of-the-art models on three benchmark datasets.

The rest of this article is organized as follows. Section II gives a brief review of the most related work. In Section III, we present the details of the proposed method. Section IV provides a series of experimental results, discussion, and model analysis. In Section V, we give a conclusion of this article.

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II. Related Work

A. Traditional RSCD Methods

The early remote sensing images have low or middle resolution and only contain several diverse objects, i.e., buildings, trees, and roads, due to the limitation of the sensor technique. In the past few decades, many transformation-based [28], [29] and image-algebra-based [30], [31], [32], [33], [34] RSCD methods have been proposed to perform change detection on these images. For the image-algebra-based approaches, they usually use image differencing [30], image regression [31], [32], [33], [34], or change vector analysis [35] to generate the difference images (DIs), in which a suitable threshold needs to be chosen to decide the changed regions. The transformation-based methods exploit some transform techniques, i.e., principal component analysis (PCA) [28] and tasseled cap transformation [29], to transform the remote sensing images into other spaces, in which the change information of the bitemporal images is enhanced. The above methods obtain change cues mainly based on hand-crafted features. However, hand-crafted features are of low robustness and insufficient semantics, which are only fit for low- or middle-resolution RSCD tasks. With the development of sensor techniques, high-resolution (HR) and very HR (VHR) remote sensing images are used commonly to obtain in many real-world applications. For HR and VHR remote sensing images, the texture of ground objects is more complex and abundant, which makes it more challenging to identify the changes.

B. CNN-Based RSCD Methods

With the development of deep CNNs, many CNN-based methods have been proposed and become the dominant solutions for the RSCD task in recent years [15], [17], [36], [37], [38], [39], [40], [41], [42], [43], [44]. Most of them treat RSCD as a binary segmentation task, which detects the changes by classifying the extracted bitemporal features into different categories. As widely acknowledged, the extracted high-level bitemporal features in the top layers of backbones effectively locate the changed objects, while the low-level bitemporal features in the bottom layers of backbones work well in capturing the details of changed objects. Many efforts have been devoted to designing advanced network architectures to perform multilevel feature fusion for achieving promising change detection performance recently. The U-shaped architecture [45], e.g., the encoder–decoder structure with nested connections, is widely used in the existing semantic segmentation methods with promising results. Motivated by the superiority of U-net, Daudt et al. [46] proposed three different U-net-based models, i.e., FC-early fusion (EF), FC-Siam-conc, and FC-Siam-diff. Based on three different U-net-based models, Daudt et al. [46] further explored the effectiveness of two different bitemporal image fusion strategies and that of two skip connection manners for the RSCD task. UNet++ [36] extracts multiscale features by applying dense skip connections to reduce some pseudochanges. L-Unet [16] combines skip connection with long short-term memory (LSTM) to capture distinguishable temporal features for the change detection task. DARNet [47] uses dense skip connection and hybrid attention to aggregate multilevel feature representations. Besides, it progressively refines predicted change maps from high levels to low levels. More inspiring related works can refer to the recent survey [15].

Previous methods design complex network architectures to extract and aggregate multilevel bitemporal features for accuracy change detection. However, the high computation costs and large model sizes of the complex network architectures inevitably prevented the RSCD methods from being applied to some real-world applications, especially for deploying the advanced RSCD methods on UAV or satellites in the real world, with limited energy overhead and computational capability. Hence, we propose to design a lightweight change detection network with fewer parameters, lower computation costs, and promising detection accuracy for the RSCD task. To ensure lightweight, our method uses mobile networks to extract multilevel bitemporal features. NAM is then applied to strengthen the representation capability of bitemporal features with few parameters. Finally, a simple coarse-to-fine decoder with PCIM and SAM is proposed to efficiently aggregate multilevel temporal difference information for change detection.

C. Lightweight Backbone Networks

Recently, mobile vision applications, such as autonomous driving vehicles, robots, and smartphones, have attracted much attention and generated urgent demands for lightweight CNNs. These scenarios only have limited computational resources, so VGG [24] and ResNet [25] are unsuitable for such cases. To this end, a fewer efficient networks, like MobileNets [21], ShuffleNets [48], and EfficientNet [49], are proposed for the image classification task. With the efficiency and effectiveness of MobileNets, ShuffleNets, and EfficientNet, they are used to extract features for semantic segmentation [22], object detection [50], and image retrieval [51], respectively. In this article, we propose to adopt MobileNetV2 [21] as the backbone to extract bitemporal features and ensure change detection accuracy in a lightweight network setting.

III. Proposed Method

In this section, we first present an overview of the proposed method. Then, we give the details of the proposed NAM, PCIM, and SAM. Finally, we provide the hybrid loss fusion.

A. Overview

The proposed network is built upon an encoder–decoder architecture, as given in Fig. 2.

1) Encoder: As discussed above, we adopt a lightweight backbone, MobileNetV2 [21], to extract bitemporal features. The global average pooling layer and the last fully connected layer are removed from the backbone to adapt to the RSCD task. Each stage of the backbone contains a convolutional layer with a stride of 2. Hence, the feature maps of each stage are downsampled into a half resolution of the previous stage. Given a registered pair of images $\boldsymbol{t}^1$ and $\boldsymbol{t}^2$, the output bitemporal featured for five stages can be denoted as $\boldsymbol{f}_1^1, \boldsymbol{f}_1^2, \ldots, \boldsymbol{f}_5^1, \boldsymbol{f}_5^2$, respectively. To reduce the computation

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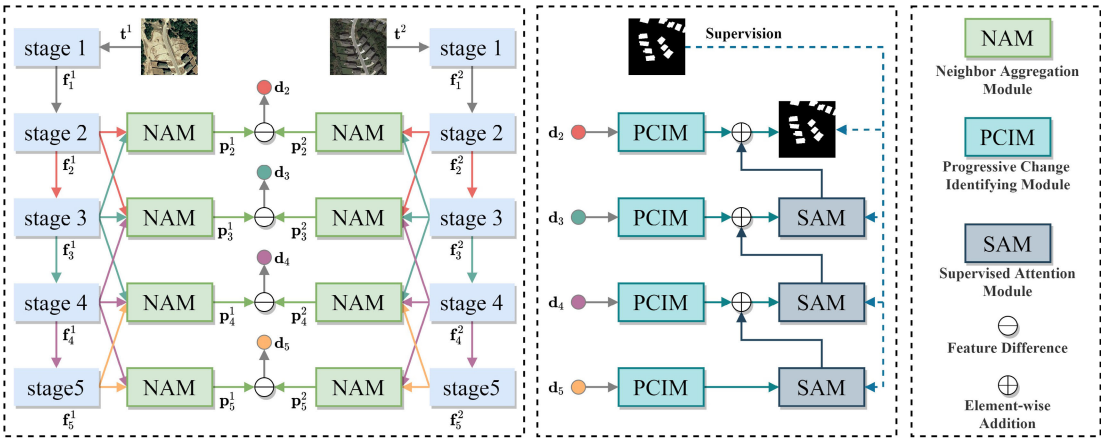


Fig. 2. Framework of the proposed A2Net. The temporal features are extracted from a registered pair of images by weight-shared MobileNetV2 [21]. Then, we use NAM to merge the temporal features within neighbor stages of the backbone to enhance their feature representation capability. PCIM is designed to capture the temporal change information from bitemporal features at their corresponding feature levels. We stack SAM on each fusion of low-level and high-level features to polish the details of changed objects. Finally, a change map is obtained by gradually aggregating temporal difference features.

costs, we only use the feature of the last four stages to perform RSCD.

We propose an NAM to enhance the representation capability of bitemporal features. By applying NAM, the semantic information and fine-grained details of objects are further strengthened in the high-level and low-level features, respectively. This procedure can be formulated as

$$\boldsymbol{p}_i^t = \mathcal{N}\left(\boldsymbol{f}_i^t\right), \quad t \in 1, 2, \ i \in 2, 3, 4, 5 \tag{1}$$

where $\mathcal{N}(\cdot)$ represents the function of NAM, and $\boldsymbol{p}_i^t$ denotes the enhanced temporal features. The architecture of the NAM will be introduced in Section III-B. Afterward, we use a feature difference manner with an elementwise subtraction operation and a subsequent absolution operation to generate temporal difference features $\boldsymbol{d}_2, \ldots, \boldsymbol{d}_5$ from bitemporal features as

$$\boldsymbol{d}_i = \left|\boldsymbol{p}_i^1 \ominus \boldsymbol{p}_i^2\right|, \quad i \in 2, 3, 4, 5 \tag{2}$$

where $|\cdot|$ and $\ominus$ denote the absolution operation and elementwise subtraction operation, respectively.

2) Decoder: In this article, we design a lightweight decoder to perform top-down multilevel feature fusion for predicting change maps with fine details. As two basic units, PCIM and SAM are constructed to fully explore the temporal change information and reweight the features for effective multilevel feature aggregation, respectively. Details of the PCIM and SAM will be, respectively, in Section III-C and III-D. Our decoder can be formulated as

$$\begin{aligned} \tilde{\boldsymbol{d}}_{i+1} &= \mathrm{Up}\{\mathcal{P}(\mathcal{S}(\boldsymbol{d}_{i+1}))\} \\ \boldsymbol{d}_{i+1} &= \mathrm{Conv}_{3\times3}\left\{\tilde{\boldsymbol{d}}_{i+1} + \mathcal{P}(\boldsymbol{d}_i)\right\} \end{aligned} \tag{3}$$

where $\mathrm{Conv}_{3\times3}(\cdot)$ denotes a 3×3 convolutional layer with a batch normalization and a relu activation function, $\mathrm{Up}(\cdot)$ is a bilinear upsampling operation. $\mathcal{P}(\cdot)$ and $\mathcal{S}(\cdot)$ represent the functions of PCIM and SAM, respectively. In this way, the multilevel temporal difference features are elegantly aggregated for generating change maps.

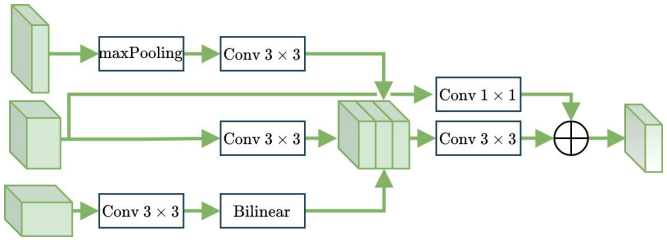


Fig. 3. Illustration of the proposed NAM.

B. Neighbor Aggregation Module

Compared with some cumbersome backbone, i.e., ResNet50 [25] and VGG19 [24], MobileNetV2 [21] is less powerful in feature representation. It is widely accepted that the high-level features contain semantic information which is beneficial to locating objects, while the low-level features consist of fine-grained boundary and texture information of objects [52], [53], [54]. Considering that the extracted multilevel features depict different aspects of the objects, we tend to integrate multilevel features to enhance the representation capability of extracted temporal features as: 1) The detailed information and semantic information can be further strengthened in low-level and high-level features, respectively and 2) the details and semantic information can be mined in mid-level features. The proposed NAM is given in Fig. 3.

The NAM merges the features extracted from nearby stages using a scheme of residual learning. As shown in Fig. 3, the inputs consist of features from the second, third, and fourth stages of the backbone. Here, we take the features extracted from t^1 image as an example. $\boldsymbol{f}_2^1$ first passes through a max pooling operation to downsample into the resolution as $\boldsymbol{f}_3^1$ and a 3×3 convolutional layer with a batch normalization and a relu activation function to increase its channels. Similarly, $\boldsymbol{f}_4^1$ passes through a 3×3 convolutional layer to reduce its channels and a bilinear upsample operation to resize the resolution as $\boldsymbol{f}_3^1$. We use a 3×3 convolutional layer to adjust

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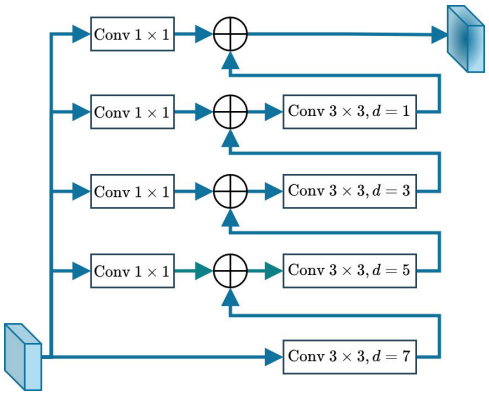


Fig. 4. Illustration of the proposed PCIM.

the channels of $\boldsymbol{f}_3^1$. The whole feature transition procedure can be denoted as

$$\begin{aligned}\boldsymbol{b}_2^1 &= \text{Conv}_{3\times3}\{\text{MP}(\boldsymbol{f}_2^1)\}\\ \boldsymbol{b}_3^1 &= \text{Conv}_{3\times3}(\boldsymbol{f}_3^1)\\ \boldsymbol{b}_4^1 &= \text{Up}\{(\text{Conv}_{3\times3}(\boldsymbol{f}_4^1)\}\end{aligned} \tag{4}$$

where MP$(\cdot)$ denotes a max pooling operation.

After feature transition, $\boldsymbol{b}_2^1$, $\boldsymbol{b}_3^1$, and $\boldsymbol{b}_4^1$ are further merged by a feature concatenation operation to aggregate multilevel information. Then, we adopt residual learning to preserve the information of $\boldsymbol{f}_3^1$ and ensure that $\boldsymbol{f}_2^1$ and $\boldsymbol{f}_3^1$ only serve as supplements. To this end, a 1×1 convolutional layer and a 3×3 convolutional layer are used to adjust the channels of $\boldsymbol{f}_3^1$ and concatenated features, respectively. The process can be formulated as

$$\boldsymbol{p}_3^2 = \text{Conv}_{1\times1}(\boldsymbol{f}_3^1) + \text{Conv}_{3\times3}\{\text{Cat}(\boldsymbol{b}_2^1, \boldsymbol{b}_3^1, \boldsymbol{b}_4^1)\} \tag{5}$$

where Cat$(\cdot)$ represents a feature concatenation operation. The NAM for other stages can be similarly formulated. After neighbor feature aggregation, we can derive the enhanced temporal features $\boldsymbol{p}_2^1, \boldsymbol{p}_2^2, \ldots, \boldsymbol{p}_5^1, \boldsymbol{p}_5^2$.

C. Progressive Change Identifying Module

We construct the PCIM for better capturing temporal change information, which uses a series of atrous convolutions and residual connects to perform multiscale feature learning. As shown in Fig. 4, there are four atrous convolutions with diverse atrous rates $\{7, 5, 3, 1\}$, and four 1×1 convolutions treated as residual connects. Suppose that $\boldsymbol{d}_2$ is the input of PCIM, we first use four 1×1 convolutional layers for feature transition as

$$\begin{aligned}\boldsymbol{d}_2^2 &= \text{Conv}_{1\times1}(\boldsymbol{d}_2)\\ \boldsymbol{d}_2^3 &= \text{Conv}_{1\times1}(\boldsymbol{d}_2)\\ \boldsymbol{d}_2^4 &= \text{Conv}_{1\times1}(\boldsymbol{d}_2)\\ \boldsymbol{d}_2^5 &= \text{Conv}_{1\times1}(\boldsymbol{d}_2)\end{aligned} \tag{6}$$

where $\boldsymbol{d}_2^2, \ldots, \boldsymbol{d}_2^5$ denote the features for the second to fifth branches of PCIM. Then, we perform multiscale feature learning as

$$\boldsymbol{d}_2^{2'} = \boldsymbol{d}_2^2 + \text{Conv}_{3\times3}^{a=7}(\boldsymbol{d}_2)$$

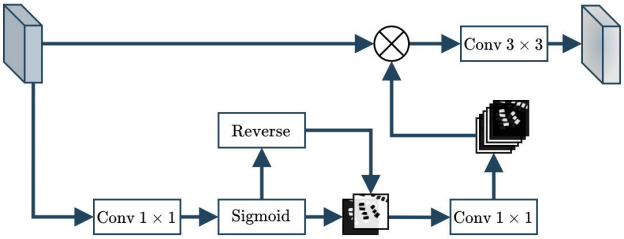


Fig. 5. Illustration of the proposed SAM.

$$\begin{aligned}\boldsymbol{d}_2^{3'} &= \boldsymbol{d}_2^3 + \text{Conv}_{3\times3}^{a=5}(\boldsymbol{d}_2^{2'})\\ \boldsymbol{d}_2^{4'} &= \boldsymbol{d}_2^4 + \text{Conv}_{3\times3}^{a=3}(\boldsymbol{d}_2^{3'})\\ \bar{\boldsymbol{d}}_2 &= \boldsymbol{d}_2^5 + \text{Conv}_{3\times3}^{a=1}(\boldsymbol{d}_2^{4'})\end{aligned} \tag{7}$$

where $\text{Conv}_{3\times3}^{a=7}(\cdot)$ denotes a 3×3 atrous convolution with an atrous rate of 7. $\bar{\boldsymbol{d}}_2$ represents the output of PCIM, i.e., $\bar{\boldsymbol{d}}_2 = \mathcal{P}(\boldsymbol{d}_2)$. In such a manner, PCIM progressively searches temporal changes from large receptive fields (with large atrous rates) to small ones (with small atrous rates) under a multibranch architecture, in which diverse branches are correlated with benefit from each other.

D. Supervised Attention Module

Feature pyramid networks [26] provide a classical architecture, i.e., coarse-to-fine, for multilevel feature fusion. However, the features in higher levels lack contextual guidance which easily leads to plenty of noise and difficulties for multilevel information aggregation. Here, we introduce an SAM to recalibrate features for better multilevel feature fusion. As shown in Fig. 5, suppose that $\bar{\boldsymbol{d}}_3$ denotes the input of SAM, we first use a 1×1 convolution layer with a sigmoid activation function to generate change map $\boldsymbol{c}_3$ as

$$\boldsymbol{c}_3 = \sigma(\text{Conv}_{1\times1}(\bar{\boldsymbol{d}}_3)) \tag{8}$$

where $\sigma(\cdot)$ is the sigmoid function, which is used to convert the change map into interval $[0, 1]$. Then, we obtain a reversed change map $\boldsymbol{c}_3^r$ by a reverse operation as $\boldsymbol{c}_3^r = \text{Rev}(\boldsymbol{c}_3)$. Rev$(\boldsymbol{c}_3)$ subtracts $\boldsymbol{c}_3$ by matrix $\boldsymbol{e}$, in which all the elements are 1. The obtained change map $\boldsymbol{c}_3$ and reversed change map $\boldsymbol{c}_3^r$ depict the contextual information of change objects and unchanged background. Here, we generate pixel attention mask $\boldsymbol{a}_3$, which has the same shape as $\bar{\boldsymbol{d}}_3$, from $\boldsymbol{c}_3$ and $\boldsymbol{c}_3^r$ using a 1×1 convolutional layer as

$$\boldsymbol{a}_3 = \text{Conv}_{1\times1}\{\text{Cat}(\boldsymbol{c}_3, \boldsymbol{c}_3^r)\}. \tag{9}$$

Afterward, we recalibrate the features $\bar{\boldsymbol{d}}_3$ with $\boldsymbol{a}_3$, resulting in supervised attention-guided features. This procedure can be formulated as

$$\bar{\boldsymbol{d}}_3^r = \text{Conv}_{3\times3}(\bar{\boldsymbol{d}}_3 \otimes \boldsymbol{a}_3) \tag{10}$$

where $\otimes$ denotes an elementwise multiplication operation. In such a formulation, $\boldsymbol{a}_3$ can be regarded as a pixelwise gate for guiding the learning procedure of $\bar{\boldsymbol{d}}_3$ and only allow the information related to changed objects passing through. Finally, the supervised attention-guided features $\bar{\boldsymbol{d}}_3^r$, produced by SAM, are further fused with higher level features.

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E. Hybrid Loss Function

For the change detection task, the ratio of changed regions is much less than that of the unchanged ones in most cases, which results in a class imbalance problem. To alleviate this issue and guide the network to learn from complex scenes, we adopt a hybrid loss, including a binary cross-entropy (BCE) loss $\mathcal{L}_{\text{bce}}$ and a dice (Dice) loss $\mathcal{L}_{\text{dice}}$ [55]. The BCE loss can be formulated as

$$\mathcal{L}_{\text{bce}}(\boldsymbol{c}, \boldsymbol{g}) = \boldsymbol{g} \cdot \log \boldsymbol{c} + (1 - \boldsymbol{g}) \cdot \log(1 - \boldsymbol{c}) \tag{11}$$

where $\cdot$ denotes a dot-product operation, $\boldsymbol{c}$ and $\boldsymbol{g}$ are the predicted change map and corresponding ground truth, respectively. The Dice loss can be formulated as

$$\mathcal{L}_{\text{dice}}(\boldsymbol{c}, \boldsymbol{g}) = 1 - \frac{2 \cdot \boldsymbol{g} \cdot \boldsymbol{c}}{\|\boldsymbol{g}\| + \|\boldsymbol{c}\|} \tag{12}$$

where $\|\cdot\|$ denotes the ℓ_1 norm. The total loss is represented as

$$\mathcal{L}(\boldsymbol{c}, \boldsymbol{g}) = \sum_{i=2}^{5} \mathcal{L}_{\text{bce}}(\boldsymbol{c}_i, \boldsymbol{g}) + \mathcal{L}_{\text{dice}}(\boldsymbol{c}_i, \boldsymbol{g}) \tag{13}$$

where $\boldsymbol{c}_i, i \in 2, 3, 4, 5$ are the predicted change maps for four stages, respectively.

IV. Experiments

A. Experimental Setup

1) Implementation Details: The proposed A2Net is implemented in the PyTorch toolbox [56] and trained/inferred on a single Nvidia Titan V GPU. In training, the backbone of A2Net is initialized with parameters of MobileNetV2 [21] pretrained on ImageNet, while the parameters of the rest part are initialized with the strategy of [57]. We use Adam optimizer [58] with a momentum of 0.9, the weight decay setting to 0.0001, and parameters β_1, β_2 as 0.9 and 0.99, respectively. The poly learning scheme is adopted so that the learning rate is adjusted as $(1 - (\text{cur_iteration}/\text{max_iteration}))^{\text{power}} \times \text{lr}$, where power and max_iteration are set as 0.9 and 40 000, respectively. We set the initial learning rate as 0.0005 and the batch size as 32. We leverage random flipping, cropping, and temporal exchanging to the input images to perform data augmentation.

B. Datasets

In our experiments, we use three RSCD benchmark datasets. The detailed information is listed as follows.

1) LEVIR [6]: This building change detection dataset consists of 637 pairs of bitemporal remote sensing images with 1024 × 1024 spatial size and 0.5-m spatial resolution, which is obtained from the Google Earth platform. Under its official dataset split setting, i.e., 7:1:2, we overlap crop images into 256 × 256 size of patches and obtain a total of 7120/1024/2048 image pairs for training/validation/test, respectively.

2) SYSU [27]: This dataset consists of 20 000 pairs of bitemporal remote sensing image patches with 256 × 256 spatial size and 0.5-m spatial resolution. The ratios of training, validation, and test are officially set to 6:2:2. This dataset contains various types of complex change scenes, including road expansion, newly built urban buildings, change of vegetation, suburban dilation, and groundwork before construction.

3) BCDD[1]*:* This is a building change detection dataset that contains a pair of images with a size of 32 507 × 15 354 and a high resolution of 0.075 m. Following BIT [19], we first overlap crop images into 256 × 256 size of patches, and then obtain a total of 6095/762/763 image pairs for training/validation/test, respectively.

4) Evaluation Metrics: We adopt five widely used evaluation metrics [59], [60], [61], namely, Kappa coefficient (κ), intersection over union (IoU), F1-score (F1), recall (Rec), and precision (Pre), to evaluate the performance of RSCD. The detailed definitions of the above evaluation metrics are formulated as follows:

$$\text{Pre} = \frac{\text{Tp}}{\text{Tp} + \text{Fp}} \tag{14}$$

$$\text{Rec} = \frac{\text{Tp}}{\text{Tp} + \text{Fn}} \tag{15}$$

$$\text{F1} = \frac{2\text{Pre} \cdot \text{Rec}}{\text{Pre} + \text{Rec}} \tag{16}$$

$$\text{OA} = \frac{\text{Tp} + \text{Tn}}{\text{Tp} + \text{Fp} + \text{Tn} + \text{Fn}} \tag{17}$$

$$\text{P}_e = \frac{(\text{Tn} + \text{Fn}) \cdot (\text{Tn} + \text{Fp}) + (\text{Fp} + \text{Tp}) \cdot (\text{Fn} + \text{Tp})}{(\text{Tp} + \text{Fp} + \text{Tn} + \text{Fn})^2} \tag{18}$$

$$\kappa = \frac{\text{OA} - \text{P}_e}{1 - \text{P}_e} \tag{19}$$

$$\text{IoU} = \frac{\text{Tp}}{\text{Tp} + \text{Fp}} \tag{20}$$

where Tp, Fp, Tn, and Fn denote the numbers of true positives, false positives, true negatives, and false negatives, respectively.

C. Comparison With the State-of-the-Art

We compare the proposed model with eight state-of-the-art RSCD approaches, including four CNN-based ones: FC-Diff [46], FC-ef [46], FC-Cat [46], STANet [6], L-Unet [16], SNUNet [17], DSIFN [18] and a transformer-based one: BIT [19]. We reproduce them using the modified codes[2] under their suggested parameters for fair comparisons. We also report the model parameters and computation costs of the above methods for reference. All the experiments are conducted with a single Nvidia Titan V GPU.

1) Quantitative Evaluation: Table I reports the quantitative evaluation comparisons of different methods in terms of κ, IoU, F1, Rec, and Pre on three RSCD datasets. We also report the size of model parameters (Params) and computation costs (FLOPs) in Table I. As seen from the results, the proposed method consistently performs better than the existing methods,

[1] http://study.rsgis.whu.edu.cn/pages/download/building_dataset.html
[2] https://github.com/Bobholamovic/CDLab

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TABLE I

QUANTITATIVE COMPARISONS IN TERMS OF κ, IOU, F1, OA, REC, AND PRE ON THREE RSCD DATASETS. THE BEST AND SECOND BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Datasets		FC-diff	FC-ef	FC-cat	STANet	L-Unet	SNUNet	DSIFN	BIT	Ours
FLOPs (G)		9.42	7.13	10.62	25.69	34.63	246.22	164.56	16.88	6.02
Params (M)		1.35	1.35	1.55	12.28	8.45	27.07	50.71	3.01	3.78
LEVIR	κ	0.8508	0.7102	0.8465	0.8839	0.8935	0.8972	0.9090	0.9007	0.9129
	IoU	0.7518	0.5679	0.7455	0.8014	0.8164	0.8220	0.8408	0.8276	0.8472
	F1	0.8583	0.7244	0.8542	0.8898	0.8989	0.9023	0.9135	0.9057	0.9173
	Rec	0.8459	0.6980	0.8453	0.8761	0.8915	0.8897	0.8944	0.8940	0.9059
	Pre	0.8712	0.7529	0.8634	0.9038	0.9064	0.9154	0.9335	0.9177	0.9290
BCDD	κ	0.8301	0.7579	0.8008	0.8941	0.9138	0.8960	0.9422	0.9356	0.9515
	IoU	0.7194	0.6219	0.6790	0.8156	0.8473	0.8186	0.8951	0.8838	0.9113
	F1	0.8368	0.7669	0.8088	0.8984	0.9173	0.9003	0.9446	0.9383	0.9536
	Rec	0.7776	0.6801	0.7646	0.8615	0.8911	0.8664	0.9388	0.9203	0.9430
	Pre	0.9057	0.8790	0.8585	0.9387	0.9452	0.9368	0.9505	0.9570	0.9644
SYSU	κ	0.6142	0.6745	0.7206	0.7100	0.7398	0.7391	0.7205	0.7333	0.7916
	IoU	0.5206	0.6011	0.6475	0.6332	0.6662	0.6673	0.6442	0.6584	0.7237
	F1	0.6847	0.7508	0.7860	0.7754	0.7996	0.8004	0.7836	0.7940	0.8397
	Rec	0.5589	0.7463	0.7800	0.7430	0.7808	0.7979	0.7511	0.7668	0.8224
	Pre	0.8836	0.7554	0.7921	0.8108	0.8195	0.8030	0.8190	0.8232	0.8577

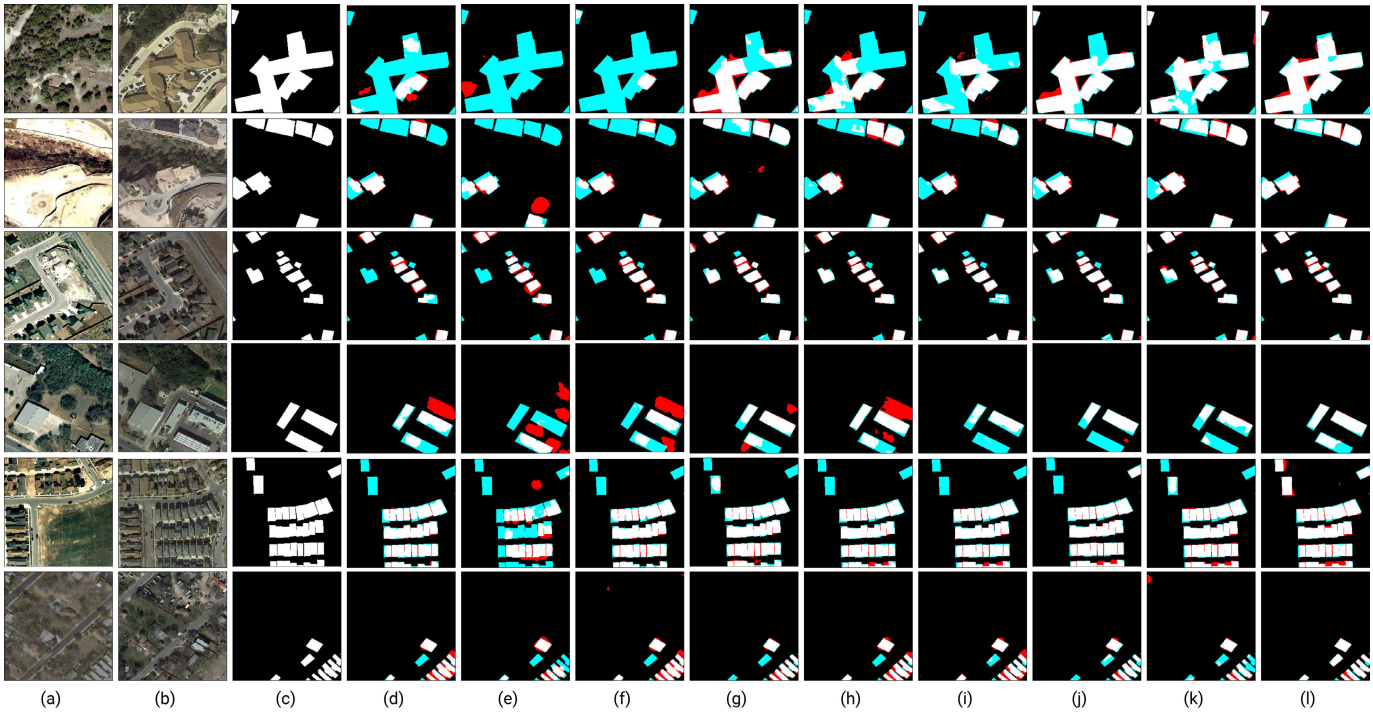


Fig. 6. Visual comparisons of the proposed method and the state-of-the-art approaches on the BCDD dataset. (a) t_1 images. (b) t_2 images. (c) Ground truth. (d) FC-Diff. (e) FC-ef. (f) FC-cat. (g) STANet. (h) L-Unet. (i) SNUNet. (j) DSIFN. (k) BIT. (l) Ours. The rendered colors represent true positives (white), false positives (red), true negatives (black), and false negatives (blue).

such as the proposed method improves 5.18%, 5.64%, and 3.95% on the SYSU-CD dataset in terms of κ, IoU, and F1, respectively. The proposed method also has a relatively small number of parameters and lowest computation costs. Taking the LEVIR dataset as an example, the DSIFN method costs 50.71 M parameters and 164.56 G FLOPs to obtain 0.9135 measured by F1, while the proposed method only costs 3.78 M (13.42× fewer) parameters and 6.02 G (27.34× lower) FLOPs to obtain 0.9173 measured by F1. Compared with the lightweight methods, i.e., BIT, FC-Diff, FC-ef, and FC-cat, the proposed method achieves significantly better detection performance than them. All these measures strongly indicate the effectiveness and superiority of the proposed lightweight A2Net method.

2) Qualitative Evaluation: The visual comparisons of different methods on three RSCD datasets are shown in

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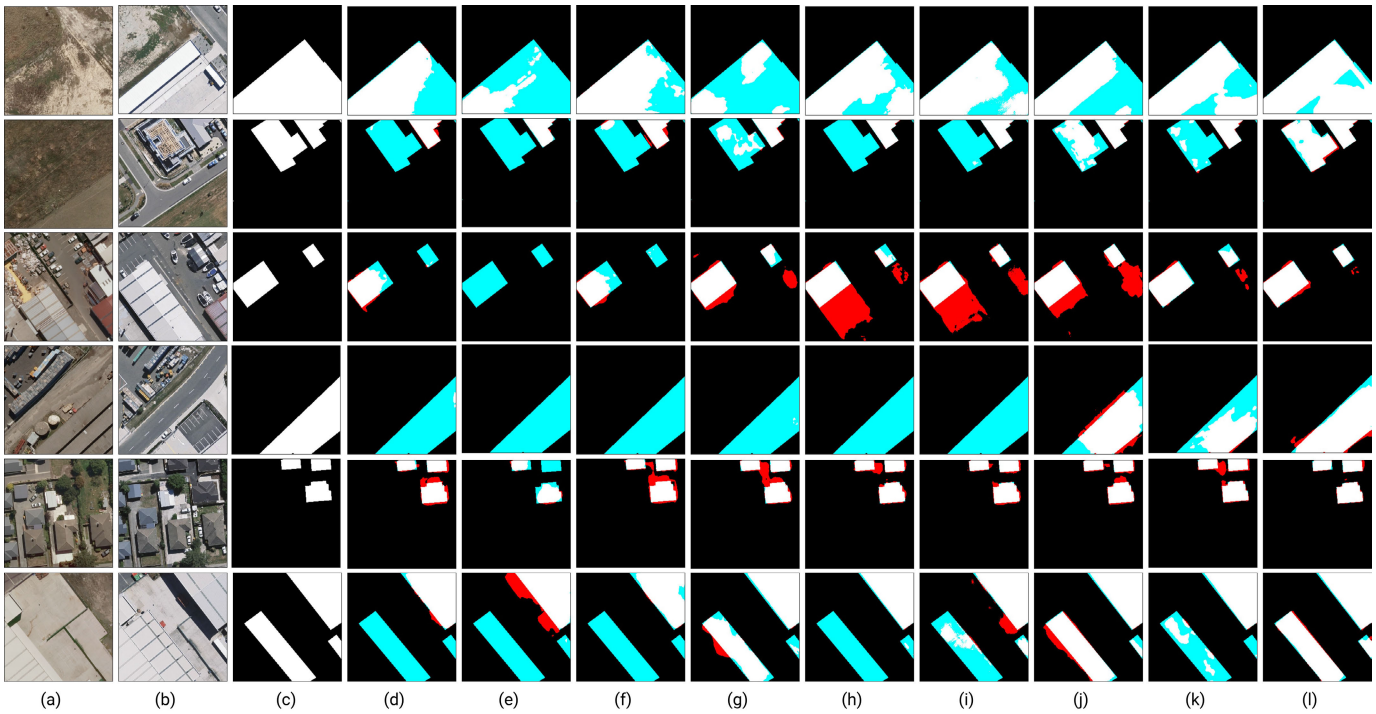


Fig. 7. Visual comparisons of the proposed method and the state-of-the-art approaches on the BCDD dataset. (a) t_1 images. (b) t_2 images. (c) Ground truth. (d) FC-Diff. (e) FC-ef. (f) FC-cat. (g) STANet. (h) L-Unet. (i) SNUNet. (j) DSIFN. (k) BIT. (l) Ours. The rendered colors represent true positives (white), false positives (red), true negatives (black), and false negatives (blue).

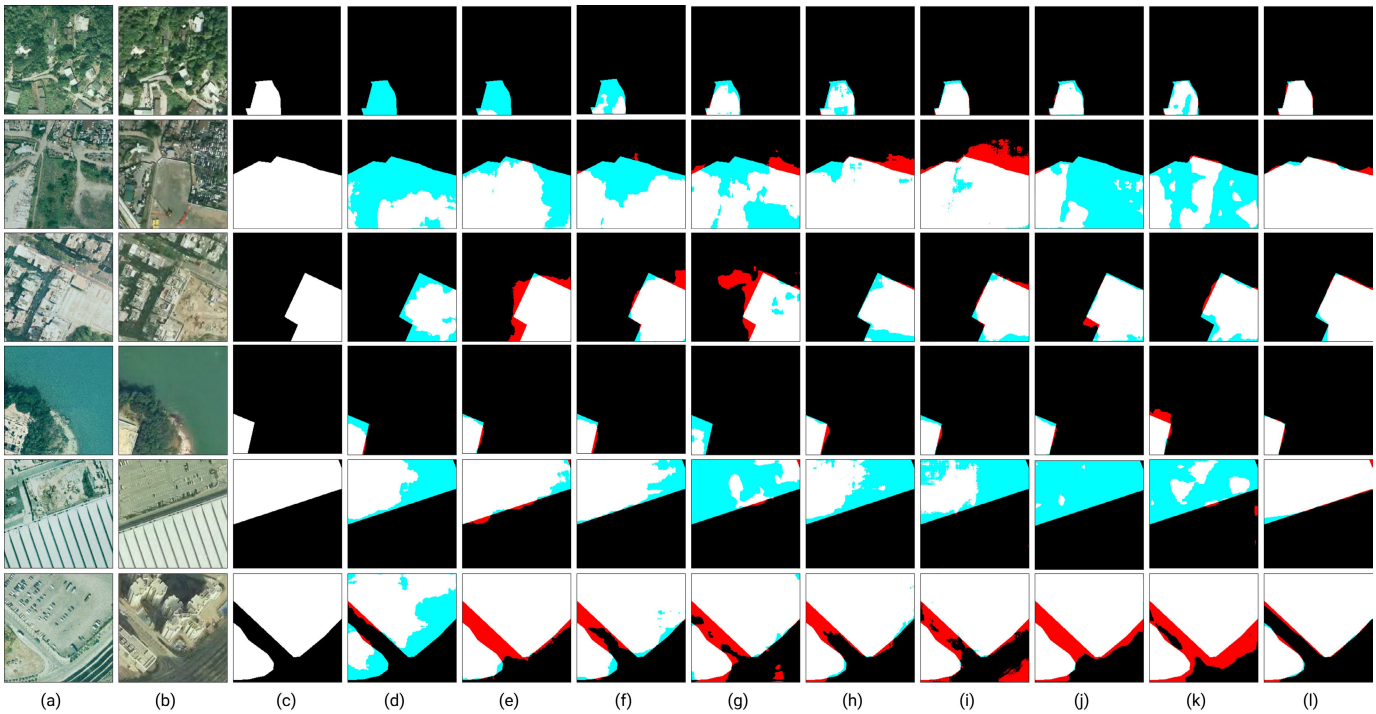


Fig. 8. Visual comparisons of the proposed method and the state-of-the-art approaches on the BCDD dataset. (a) t_1 images. (b) t_2 images. (c) Ground truth. (d) FC-Diff. (e) FC-ef. (f) FC-cat. (g) STANet. (h) L-Unet. (i) SNUNet. (j) DSIFN. (k) BIT. (l) Ours. The rendered colors represent true positives (white), false positives (red), true negatives (black), and false negatives (blue).

Figs. 6–8. For better visualization, we adopt several colors to represent true positives (white), false positives (red), true negatives (black), and false negatives (blue). From the results, we observe that the proposed method exhibits superiority in the following aspects.

1) *Advantages in Location Accuracy:* Compared with other methods, the proposed method can more accurately locate the changed objects and performs well in suppressing the interference of background in bitemporal images. In the first row of Fig. 6, FC-Diff, FC-ef, FC-cat,

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TABLE II
QUANTITATIVE COMPARISONS OF THE PROPOSED METHOD WITH DIVERSE SETTINGS IN TERMS OF κ, IOU, AND F1 ON THREE RSCD DATASETS

No.	Variants	Efficiency		LEVIR			BCDD			SYSU		
		Params (M)	FLOPs (G)	κ	IoU	F1	κ	IoU	F1	κ	IoU	F1
#01	A2Net	3.78	6.02	0.9129	0.8472	0.9173	0.9515	0.9113	0.9536	0.7916	0.7237	0.8397
(a) Neighbor Aggregation Module (NAM)												
#02	w/o NAM	3.14	3.98	0.9068	0.8374	0.9115	0.9402	0.8917	0.9427	0.7716	0.6995	0.8232
(b) Progressive Change Identifying Module (PCIM)												
#03	w/o PCIM	3.27	4.63	0.9101	0.8426	0.9146	0.9439	0.8981	0.9463	0.7780	0.7084	0.8293
#04	$\{1, 3, 5, 7\}$	3.78	6.02	0.9101	0.8427	0.9146	0.9396	0.8906	0.9422	0.7780	0.7065	0.8280
#05	$\{1, 1, 1, 1\}$	3.78	6.02	0.9126	0.8467	0.9170	0.9364	0.8853	0.9392	0.7836	0.7140	0.8331
(c) Supervised Attention Module (SAM)												
#06	w/o SAM	3.67	5.92	0.9089	0.8408	0.9135	0.9476	0.9045	0.9499	0.7732	0.7047	0.8268
#07	w/o change	3.78	6.02	0.9077	0.8389	0.9124	0.9517	0.9116	0.9537	0.7768	0.7046	0.8267
#08	w/o unchange	3.78	6.02	0.9099	0.8424	0.9145	0.9472	0.9037	0.9494	0.7872	0.7176	0.8356
(d) Different Backbones												
#09	ResNet18	17.50	28.90	0.9112	0.8444	0.9156	0.9517	0.9115	0.9537	0.7812	0.7131	0.8325
#10	ResNet50	34.18	47.40	0.9146	0.8500	0.9189	0.9552	0.9177	0.9571	0.7814	0.7120	0.8317
(e) Loss Functions												
#11	BCE	3.78	6.02	0.9064	0.8367	0.9111	0.9397	0.8908	0.9422	0.7774	0.7057	0.8275
#12	Dice	3.78	6.02	0.9104	0.8432	0.9149	0.9475	0.9043	0.9498	0.7791	0.7076	0.8288

STANet, L-Unet, SNUNet, and DSIFN can hardly completely locate the changed buildings, resulting in wrong prediction. In contrast, the changed buildings are accurately and completely identified by the proposed method.

2) *Advantages in Distinguishing Pseudochanges:* In the third row of Fig. 7, the same buildings show different colors at different times which causes some irrelevant changes contrary to real building changes. Many methods, e.g., STANet, L-Unet, SNUNet, DSIFN, and BIT, fail to exclude these changes, while the proposed method can well handle the pseudochanges resulting from land cover appearance diversity. In the sixth row of Fig. 8, many methods cannot distinguish the pseudochanges in roads that contain a similar appearance to buildings. On the contrary, the proposed method can well identify the boundary and body of the building changes.

3) *Advantages in Content Integrity:* In Fig. 8, the complex sense changes cannot be entirely identified by these compared methods. The proposed method can achieve relatively intact detection results. The main reasons are that first, the proposed method fuses temporal features within nearby stages of the backbone. Second, the proposed method identifies temporal changes at multiple levels and generates multilevel temporal difference feature representations which capture the rich contextual information at diverse levels. Third, the proposed method gradually aggregates multilevel temporal difference features under the guidance of contextual attention in a coarse-to-fine strategy to provide a finer change map for the RSCD task.

D. Ablation Studies

To verify the effectiveness of the components and configurations of the proposed A2Net, we conduct comprehensive ablation studies on three RSCD datasets.

1) Effectiveness of NAM: The NAM aims to enhance the representation capability of the extracted bitemporal features by fusing the information of neighbor stages within the backbone. To validate the effectiveness of NAM, we replace it with a 3×3 convolutional layer that has the same output channels as NAM and formulate a method termed wo-NAM (#02 in Table II). We report the quantitative comparison results in Table II. As can be seen, A2Net outperforms A2Net-wo-NAM, showing that aggregating features of neighbor stages is significant in A2Net.

2) Effectiveness of PCIM: In the proposed A2Net, we use PCIM to progressively capture the temporal change information of objects. To verify its effectiveness, we first replace it with a 3×3 convolutional layer of which the output channels are set the same as PCIM, to construct a wo-PCIM (#03 in Table II), and report the quantitative comparisons results in Table II. From the result, when PCIM is replaced, the detection accuracy is degraded about 1.36%, 1.53%, and 1.04% in terms of κ, IoU, and F1 on SYSU, respectively. This demonstrates the effectiveness of the proposed PCIM. Furthermore, we show the results of different settings of atrous rate for PCIM, e.g., $\{1, 3, 5, 7\}$ and $\{1, 1, 1, 1\}$ in Table II. We can observe that the original setting $\{7, 5, 3, 1\}$ can obtain the best performance, demonstrating the effectiveness of progressively searching temporal changes from large receptive fields (with large atrous rates) to small ones (with small atrous rates).

3) Effectiveness of Supervised Attention Module (SAM): We apply SAM to reweight features for better conducting the coarse-to-fine feature aggregation. To validate its effectiveness, we perform an ablation study by replacing SAM with 3×3 convolutions of which the output channels are set the same as SAM and term as w/o SAM (#06 in Table II). From the results in Table II, we observe that A2Net outperforms wo-SAM significantly, suggesting the effectiveness of the proposed SAM. To further study the settings of supervised

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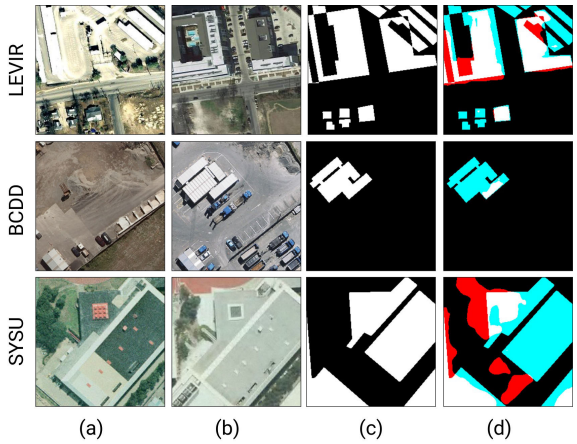


Fig. 9. Visual of some failure cases of the proposed A2Net on three RSCD datasets. (a) t_1 images. (b) t_2 images. (c) Ground truth. (d) Ours. The rendered colors represent true positives (white), false positives (red), true negatives (black), and false negatives (blue).

attention, we construct two methods by only using the changed mask or unchanged mask for formulating the pixel attention. From the results in Table II, we can obverse that only using a changed mask or unchanged mask has worse performance than the default setting. Therefore, the default strategy is the best choice.

4) Discussions of Different Backbones: We further validate the proposed method with different backbones, e.g., ResNet18 and ResNet50. From the results in Table II, we observe that A2Net with MobileNetV2 outperforms better than that with ResNet18 and slightly performs worse than that with ResNet50 on the LEVIR and BCDD datasets. As a result, the proposed methods with MobileNetV2 can achieve satisfactory performance as that with ResNet18 and ResNet50 backbones, indicating the effectiveness of a lightweight backbone.

5) Discussions of the Loss Function: In this article, we adopt the hybrid loss consisting of the Dice loss and BCE loss. To validate its effectiveness, we conduct experiments on the methods training with only BCE loss or only Dice loss in Table II. As can be seen, the method with Dice loss performs better than that with BCE loss. Besides, only using Dice loss or BCE loss has worse detection performance than the hybrid loss. There, we use the hybrid of the Dice loss and BCE loss as our model training loss.

E. Discussions About Failure Cases

Although the proposed A2Net can achieve progressing performance for RSCD in a global view, there are some challenging cases needed to be addressed for further improvement. In Fig. 9, we present some representative failure cases of three RSCD datasets. In the above failure cases, the proposed A2Net predicts the wrong changed regions or cannot capture the changed regions due to the changed objects showing fewer differences with the background. Even so, the improvement as discussed in Sections IV-C1 and IV-C2 verifies that the proposed A2Net can copy with most challenging scenarios well compared with the state-of-the-art methods for the RSCD tasks.

V. Conclusion

In this work, we proposed a novel lightweight RSCD network via progressive feature aggregation and supervised attention. We introduce an NAM to strengthen the less power temporal features extracted by mobile backbone networks. Then, we propose a PCIM to capture the temporal change information for bitemporal features at diverse feature levels. Finally, we use an SAM to reweight the features for better aggregating multilevel information in a coarse-to-fine scheme. The experimental results obtained on three high spatial resolution RSCD datasets indicate that the proposed method is superior to the state-of-the-art change detection methods. This work could be served as a strong baseline for lightweight RSCD.

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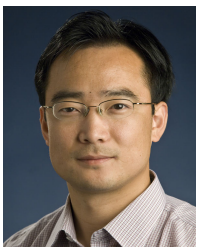
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